

A classifier produced the following confusion matrix for a binary classification problem:

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

Calculate the following measures:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

$$\text{Accuracy} = (TP + TN) / \text{Total}$$

$$= (50 + 35) / 100 = 85 / 100 = \mathbf{0.85 = 85.00\%}$$

$$\text{Precision} = TP / (TP + FP)$$

$$= 50 / (50 + 5) = 50 / 55 = (\text{divide by 5}) 10 / 11 \approx \mathbf{0.9090909 = 90.91\%}$$

$$\text{Recall (Sensitivity)} = TP / (TP + FN)$$

$$= 50 / (50 + 10) = 50 / 60 = (\text{divide by 10}) 5 / 6 \approx \mathbf{0.8333333 = 83.33\%}$$

$$\text{F1-score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Using fractions: Precision = 10/11, Recall = 5/6

$$\text{Precision} * \text{Recall} = (10/11)(5/6) = 50/66 = 25/33$$

$$\text{Precision} + \text{Recall} = 10/11 + 5/6 = (60 + 55) / 66 = 115/66$$

$$\text{So } F1 = 2(25/33) / (115/66) = (50/33) * (66/115)$$

Simplify: $66/33 = 2 \rightarrow (50 * 2) / 115 = 100/115 = (\text{divide by 5}) 20/23 \approx \mathbf{0.8695652 = 86.96\%}$

	Predicted A	Predicted B	Predicted C	Actual Total
Actual A	40	5	5	50
Actual B	3	30	7	40
Actual C	2	8	25	35
Pred Total	45	43	37	125

Step 1: Overall Accuracy

$$\text{Accuracy} = 40 + 30 + 25 / 125 = 0.76 = 76\%$$

Step 2: Per-class Precision, Recall, F1

For Class A

- $TP = 40$
- $FP = (\text{Predicted A but not A}) = 45 - 40 = 5$
- $FN = (\text{Actual A but not predicted A}) = 50 - 40 = 10$

$$\text{Precision(A)} = 40 / (40+5) = 40/45 \approx \mathbf{0.8889}$$

$$\text{Recall(A)} = 40 / (40+10) = 40/50 = \mathbf{0.8}$$

$$F1(A) = 2 * (0.8889 * 0.8) / (0.8889 + 0.8) \approx \mathbf{0.8421}$$

For Class B

- $TP = 30$
- $FP = (\text{Predicted B but not B}) = 43 - 30 = 13$
- $FN = (\text{Actual B but not predicted B}) = 40 - 30 = 10$

$$\text{Precision(B)} = 30/43 \approx \mathbf{0.6977}$$

$$\text{Recall(B)} = 30/40 = \mathbf{0.75}$$

$$F1(B) = 2 * (0.6977 * 0.75) / (0.6977 + 0.75) \approx \mathbf{0.7234}$$

For Class C

- $TP = 25$
- $FP = (\text{Predicted C but not C}) = 37 - 25 = 12$
- $FN = (\text{Actual C but not predicted C}) = 35 - 25 = 10$

$$\text{Precision(C)} = 25/37 \approx \mathbf{0.6757}$$

$$\text{Recall(C)} = 25/35 \approx \mathbf{0.7143}$$

$$F1(C) = 2 * (0.6757 * 0.7143) / (0.6757 + 0.7143) \approx \mathbf{0.6944}$$

Step 3: Macro-averaged metrics

(average over all classes equally)

- $\text{Precision (macro)} = (0.8889 + 0.6977 + 0.6757) / 3 \approx \mathbf{0.7541}$
- $\text{Recall (macro)} = (0.8 + 0.75 + 0.7143) / 3 \approx \mathbf{0.7548}$

- $F1(\text{macro}) = (0.8421 + 0.7234 + 0.6944) / 3 \approx \mathbf{0.7533}$